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Allowing exception throwing in constant-evaluation

Introduction

Since adding the `constexpr` keyword in C++11, WG21 has gradually expanded the scope of language features for use in constant-evaluated code. At first, users couldn't even use `if`, `else`, or loops. C++14 added support for them. C++17 added support for `constexpr` lambdas. C++20 finally added the ability to use allocation, `std::vector` and `std::string`. These improvements have been widely appreciated, and lead to simpler code that doesn't need to work around differences between normal and `constexpr` C++.

The last major language feature from C++ still not present in `constexpr` code is the ability to throw an exception. This absence forces library authors to use more intrusive error reporting mechanisms. One example would be the use of `std::expected` and `std::optional`. Another one is the complete omission of error handling. This leaves users with long and confusing errors generated by the compiler.

Throwing exceptions in constant evaluated code is the preferred way of error reporting in the proposal adding [Static Reflection for C++26](#). Some meta-functions can fail, and allowing them to throw will significantly simplify reflection code.

Changes

- [R4](#) → [R5](#): Marking a note part of the wording as addition. Move wording about lifetime of exceptions and implicit copies to `expr.const p5`.
- [R3](#) → [R4](#): Added missing library exception type `bad_typeid`. Added wording to specify literal encoding of `std::exception::what()` member function's result during constant evaluation.